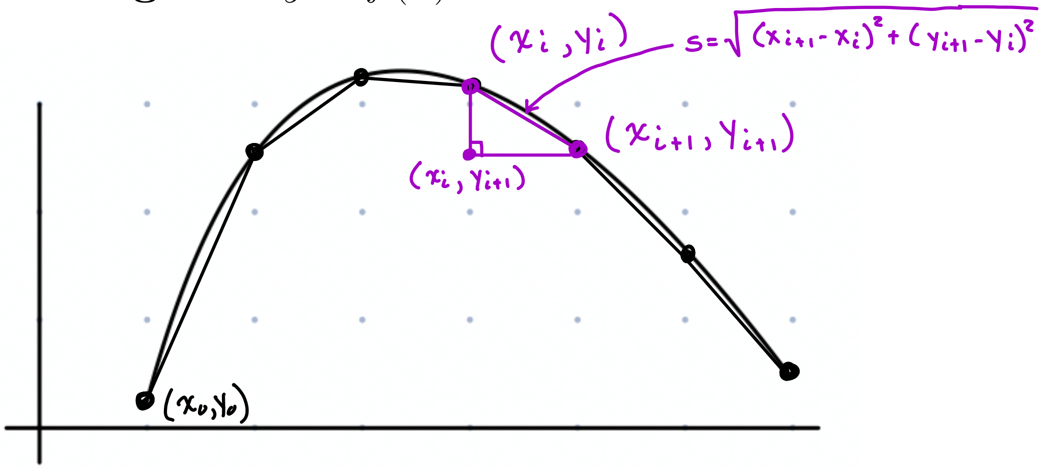
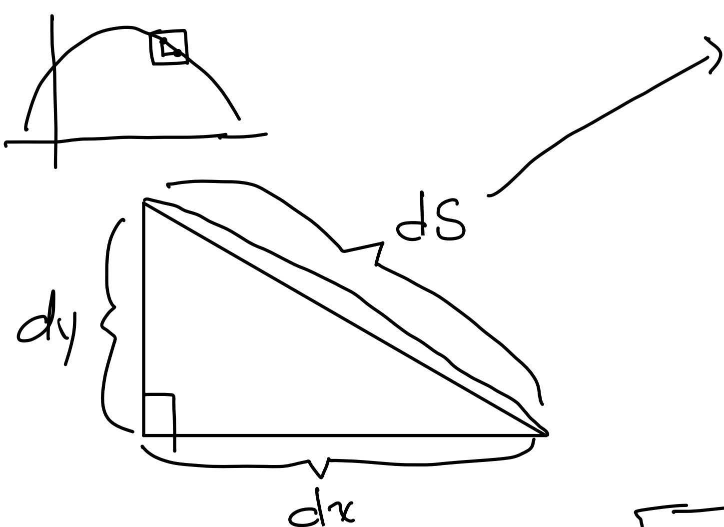
8.1 Arc Length

1 Length of $y = f(x)$



Plan: Let's approximate the length of this curve by inscribing some line segments:



$$dS^2 = (dx)^2 + (dy)^2$$

$$dS = \sqrt{(dx)^2 + (dy)^2}$$

$\frac{dx}{dx}$ leave outside / move inside

$$= \sqrt{\left[(dx)^2 + (dy)^2\right]\frac{1}{(dx)^2}}\; dx$$

$$= \sqrt{\left(\frac{dx}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^2}\; dx$$

$$\boxed{dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\; dx}$$

Memorize:

$$dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

The arc length of the curve from $x = a$ to $x = b$ is $\displaystyle\int_a^b dS$

1.0.1 Example:

Find the arc length of the function $f(x) = \frac{3}{5}x$ from $x = 0$ to $x = 10$.

$$\frac{dy}{dx} = \frac{3}{5}$$

$$dS = \sqrt{1 + \left(\frac{3}{5}\right)^2}\, dx = \sqrt{1 + \frac{9}{25}}\, dx$$

$$= \sqrt{\frac{34}{25}}\, dx$$

$$L = \int_0^{10} \frac{\sqrt{34}}{5}\, dx = \frac{\sqrt{34}}{5} x \Big]_0^{10}$$

$$= \frac{\sqrt{34}}{5} \cdot (10-0) = \boxed{2\sqrt{34}}$$

1.0.2 Example:

Find the length of the parabola $y = x^2$ from $x = 0$ to $x = 3$.

$$\frac{dy}{dx} = 2x \qquad dS = \sqrt{1 + 4x^2}\, dx$$

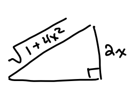
$$L = \int_0^3 \sqrt{1 + 4x^2}\, dx \rightarrow \frac{1}{2}\int \sec\theta \sec^2\theta\, d\theta = \frac{1}{2}\int \sec^3\theta\, d\theta$$

$$\boxed{\begin{aligned} 2x &= \tan\theta \\ 2dx &= \sec^2\theta\, d\theta \\ \sqrt{1+\tan^2\theta} &= \sec\theta \end{aligned}}$$

$$= \frac{1}{2}\cdot\frac{1}{2}\left(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|\right)$$

$$= \frac{1}{4}\left(2x\sqrt{1+4x^2} + \ln\left|\sqrt{1+4x^2} + 2x\right|\right)\Big]_0^3$$

$$= \frac{1}{4}\left(6\sqrt{1+36} + \ln\left|\sqrt{37}+6\right|\right) - \underbrace{\frac{1}{4}(0 + \ln 1)}_{0}$$

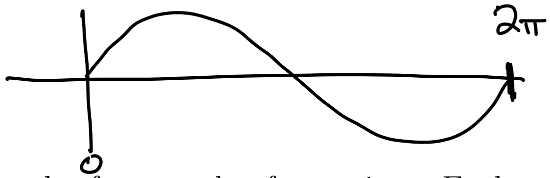


$\tan\theta = \frac{2x}{1}$

$\sec\theta = \sqrt{1+4x^2}$

$$= \boxed{\frac{1}{4}\left(6\sqrt{37} + \ln\left(\sqrt{37}+6\right)\right)}$$

1.0.3 Example:



Set up an integral for the length of one cycle of $y = \sin x$. Evaluate it using Desmos or another calculator.

$$dS = \sqrt{1 + \cos^2 x}\, dx \qquad \frac{dy}{dx} = \cos x$$

$$L = \int_0^{2\pi} \sqrt{1 + \cos^2 x}\, dx \approx 7.64$$

2 Length of $x = f(y)$

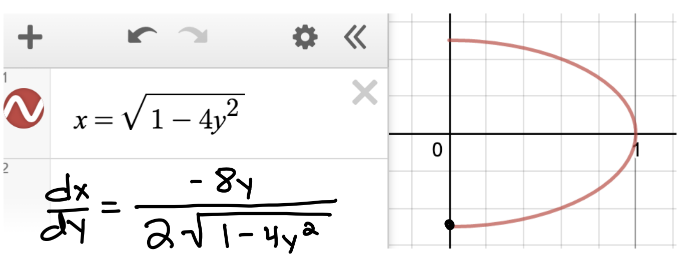
Memorize:

$$dS = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

The arc length of the curve from $y = c$ to $y = d$ is $\int_c^d dS$

2.0.1 Example:

Set up an integral for the length of the half-ellipse shown here, and evaluate it using a calculator.



$$\frac{dx}{dy} = \frac{-8y}{2\sqrt{1-4y^2}}$$

$$0 = \sqrt{1-4y^2}$$
$$0 = 1 - 4y^2$$
$$4y^2 = 1$$
$$y^2 = \frac{1}{4}$$

$$\frac{dx}{dy} = \frac{-4y}{\sqrt{1-4y^2}} \qquad dS = \sqrt{1 + \left(\frac{-4y}{\sqrt{1-4y^2}}\right)^2}\, dy$$

$$= \sqrt{1 + \frac{16y^2}{1-4y^2}}$$

$$\boxed{L = \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{1 + \frac{16y^2}{1-4y^2}}\, dy}$$

Ex: Find the length of $y = \ln(\sec x)$ from $x = 0$ to $x = \frac{\pi}{4}$

$$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x$$

$$dS = \sqrt{1 + \tan^2 x}\, dx$$

$$= \sqrt{\sec^2 x}\, dx$$

$$dS = \sec x\, dx$$

$$L = \int_0^{\pi/4} \sec x\, dx = \ln\left|\sec x + \tan x\right|\Big]_0^{\pi/4}$$

$$= \ln\Big(\underbrace{\sec\tfrac{\pi}{4}}_{\sqrt{2}} + \underbrace{\tan\tfrac{\pi}{4}}_{1}\Big) - \ln\big(\underbrace{\sec 0}_{1} + \underbrace{\tan 0}_{0}\big)$$

$$= \boxed{\ln(\sqrt{2}+1)} - \underbrace{\ln 1}_{0} \approx .881$$